

Q-Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

To troubleshoot a DNS issue where devices can access websites using IP addresses but cannot access them using domain names, and to identify and resolve the DNS configuration problem.

Step 1: Verify IP Connectivity

Open **Command Prompt** on the client PC and test connectivity using an IP address.

```
ping 8.8.8.8
```

Output:

```
Reply from 8.8.8.8: bytes=32 time=20ms TTL=117
```

This confirms that the network connection is working.

Step 2: Test DNS Resolution

Run the following command:

```
nslookup www.google.com
```

Output:

```
DNS request timed out.
```

```
*** Request to Unknown timed-out
```

```
or
```

```
*** Can't find www.google.com: Server failed
```

This indicates that the DNS server is unreachable or incorrectly configured.

Step 3: Check DNS Server Configuration

Run:

```
ipconfig /all
```

output:

IP Address.....192.168.1.2
Subnet Mask.....255.255.255.0
Default Gateway.....192.168.1.1
DNS Servers.....192.168.1.254

The DNS server address (192.168.1.254) is incorrect.

Step 4: Identify the Cause

The client received an incorrect DNS server address through DHCP or manual configuration. Because of this, domain names cannot be translated into IP addresses.

Solution

Configure the correct DNS server address.

Example:

DNS Server: 8.8.8.8

or configure the IP address of your local DNS server if one is available.

Renew the network settings or manually update the DNS configuration.

Step 5: Verify the Fix

Run:

nslookup www.google.com

Expected output:

Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name: www.google.com
Addresses: 142.250.xx.xx

Test connectivity using the domain name:

ping www.google.com

output:

Pinging www.google.com [142.250.xx.xx]

Reply from 142.250.xx.xx: bytes=32 time=25ms TTL=118

This confirms that DNS resolution is functioning correctly.

Troubleshooting Steps

1. Verify network connectivity by pinging an IP address (ping 8.8.8.8).
2. Test DNS resolution using nslookup www.google.com.
3. Check the client's DNS server configuration with ipconfig /all.
4. Compare the configured DNS server with the expected DNS server.
5. Correct the DNS server address in the DHCP settings or manually configure it.
6. Renew the IP configuration or reconnect the client.
7. Test nslookup and ping using the domain name to confirm the issue is resolved.

The client could access websites using IP addresses but failed to access them using domain names due to an incorrect DNS server configuration. After configuring the correct DNS server address, domain name resolution worked successfully, and websites could be accessed by name.